# DEEPTHA SABARISH

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## PROFESSIONAL SUMMARY

Mechanical Engineer with 3+ years of industry experience in product development, and manufacturing engineering. Proven analytical and problem-solving skills with ability to troubleshoot design and manufacturing challenges. Strong background in DFM/DFA, GD&T, testing and cross-functional collaboration in regulated and R&D environments.

## EDUCATION

**New Jersey Institute of Technology** **Aug 2024 - May 2026**
MS in Mechanical Engineering
Relevant courses - Robotics, AI, Embedded Systems, Mechanobiology

**CMR Institute of Technology** **Aug 2017 - June 2021**
B.E in Mechanical Engineering
Relevant courses – CAD, Fluid Dynamics, Thermodynamics, Mechanics

## SKILLS

Mechanical Design: SolidWorks (CSWA), Creo, CATIA, AutoCAD, Solid Edge
Manufacturing &DFM: GD&T, Machining, Welding, DFA/DFM
Prototyping & Testing: Rapid Prototyping, 3D printing, Design Validation, DOE, Root Cause Analysis
Programming & Controls: Python, C, C++, ROS, Arduino
Simulation & Analysis: MATLAB, SolidWorks
Documentation: Engineering Drawing, Test Reports, Manufacturing Documentation

## RELEVANT EXPERIENCE

**Engineering Intern, Manufacturing** **Jun 2025 – Aug 2025**
Medtronic Eatontown, New Jersey

- Redesigned the base to reduce overall weight by ~40% and tool to adjust band tension of the existing bandsaw to enhance user comfort enabling its replacement
- Investigated water retention issues after sterilization and proposed viable design adjustments and temperature-time ratios
- Supported mold-volume verification, sample separation and classification for new equipment testing
- Earned clean-room gowning certification and conducted multiple camera reviews to ensure protocol compliance
- Created and updated detailed SolidWorks models and drawings for production tooling, incorporating GD&T and manufacturability requirements.

**Engineering Consultant, R&D** **Jan 2024 – July 2024**
Xpredict Automation Solutions Bangalore, India

- Field-tested and validated system performance in real-world industrial environment
- Designed patentable product designs to reduce the contamination of electrode surface by 65% during Electrode Coagulation and the development of reactor cells as an add-on to enable Electrode Coagulation increasing efficiency by 35%
- Ideated HydRec, water reclamation system, as a domestic product
- Revised manufacturing process designs to reduce manufacturing time by 2 weeks
- Aided the design of an STP (Sewage Treatment Plant) on AutoCAD

**Junior Engineer, R&D** **June 2021 – Dec 2022**
Xpredict Automation Solutions Bangalore, India

- Worked extensively on HydRec, water reclamation system iterations, using chemical and electrode coagulation processes
- Created 3D models from 2D drawings and optimized workflows using SolidWorks and AutoCAD
- Gained hands-on experience with machining, welding, and finishing processes on shop-floor and vendor builds
- Standardized manufacturing processes; authored DFM based production and service manual
- Set up experiments to optimize the water reclamation system by tuning processing time and voltage across materials
- Reverse-engineered peristaltic pumps to gain expertise in safety protocols, product development and manufacturing techniques.

## PROJECTS & PUBLICATIONS

- Real-time gesture-recognition system using custom triple-channel EMG sensors and signal-processing algorithms (Microcontrollers-Arduino Leonardo, EMG, Machine Learning-Random Forest Algorithm) **Dec 2025**
- Domestic water reclamation systems in flow cell design and prototype development (CAD, Flow simulation-SolidWorks, DFM, R&D) **April 2024**
- "Experimental Study on the Effect of Biogas Diesel Engines Driven by Silkworm Larval Litter, Cyperus Rotundus, and Cattle Urine" https://doi.org/10.1016/j.matpr.2021.05.464 **June 2021**

## ROLES & RESPONSIBILITIES

**Teaching Assistant, NJIT** **Jan 2025 – May 2026**

- Mentored undergraduate students in 3D modeling, assemblies, drawings and GD&T and, Fluid Mechanics

**Chief Editor, CMR Institute of Technology** **Jan 2020 – June 2020**

- Led a team of four in editing the department magazine (Machinedge)